Electronics Product Engineering Workshop

ECP307

PROJECT

SMART GREENHOUSE AUTOMATION SYSTEM

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AIM

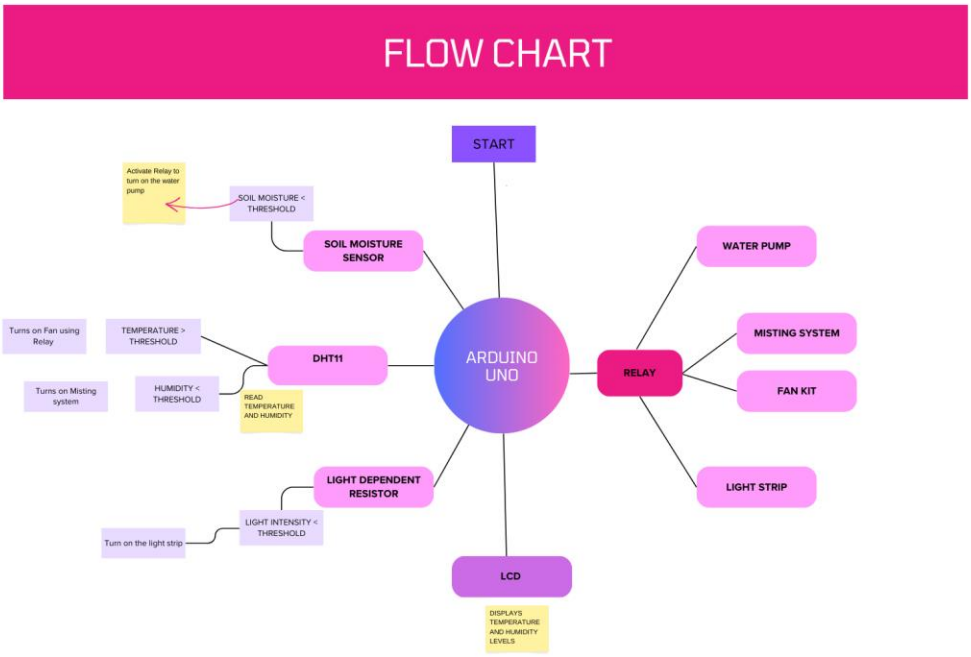
The aim of this project is to design and develop a Smart Greenhouse Automation System that optimizes plant growth by automating environmental control processes. The system will monitor and regulate key parameters such as temperature, humidity, soil moisture, and light intensity using IoT-based sensors and actuators. By integrating automation and real-time monitoring, the project aims to improve crop yield, reduce manual intervention, and enhance resource efficiency while promoting sustainable agricultural practices.

INTRODUCTION

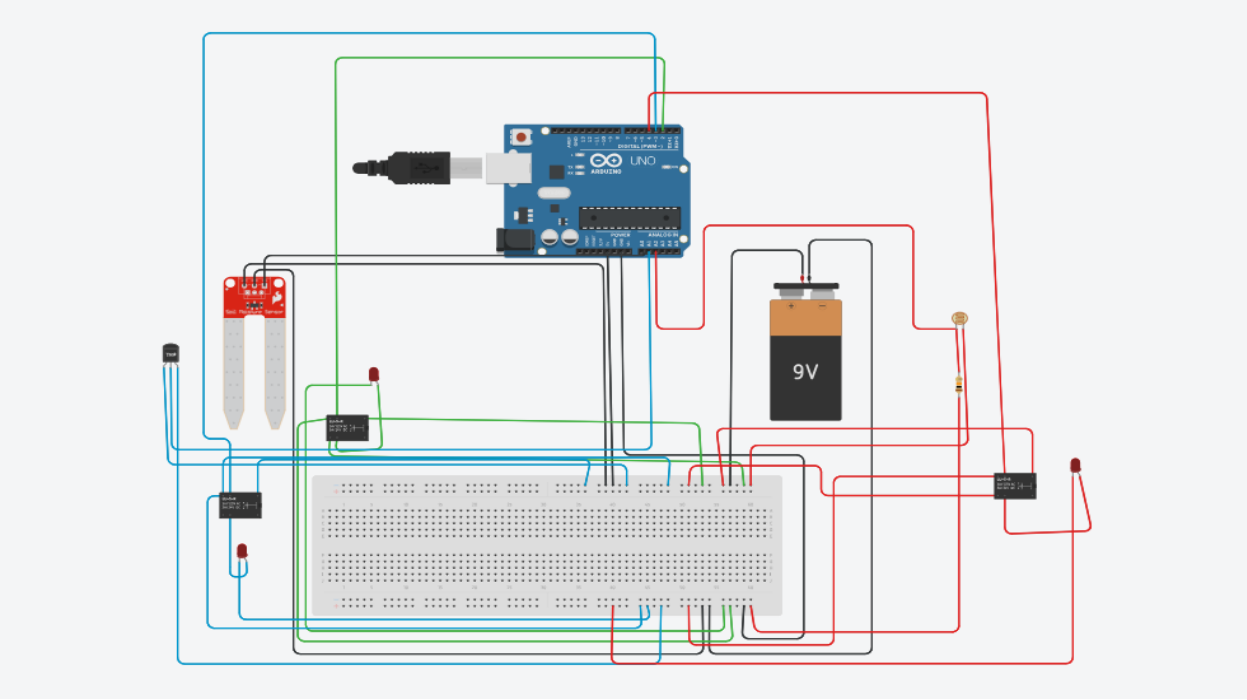
This report presents the development of a Smart Greenhouse Automation System that integrates sensors and microcontrollers to monitor and control key environmental factors like temperature, humidity, soil moisture, and light intensity. Using Arduino Uno, sensors, and actuators, the system ensures optimal plant growth with minimal human intervention. It allows real-time monitoring and control. This report covers the design, implementation, and benefits of the system, highlighting its role in modernizing agriculture through automation.

METHODOLOGY

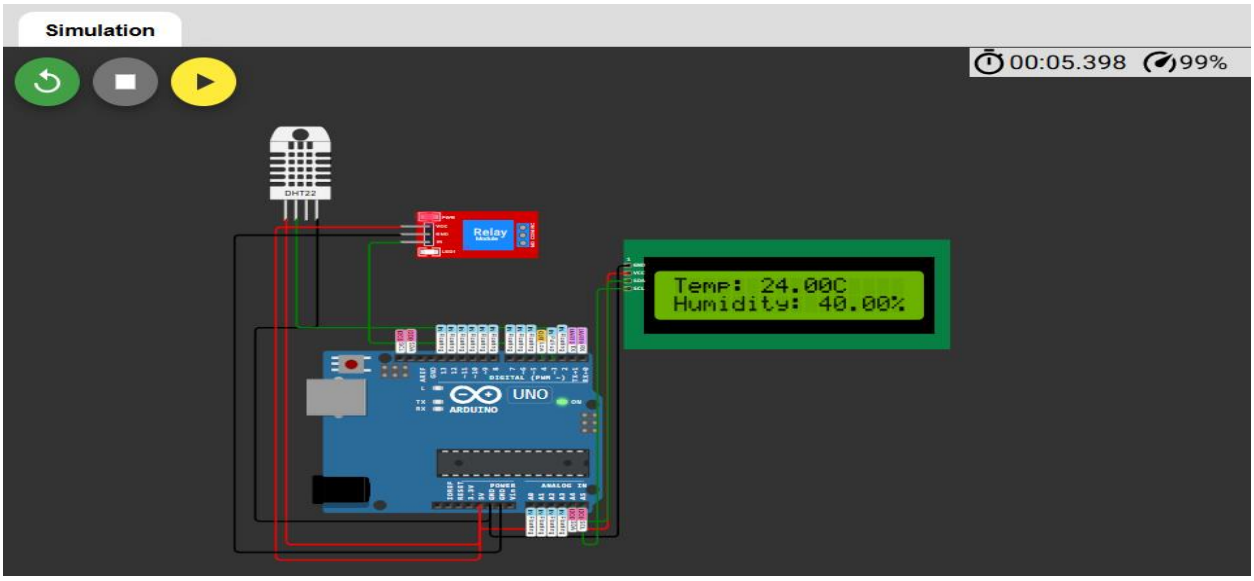
The methodology involves sensor data collection, processing, and automated control to maintain optimal greenhouse conditions. The Data is collection is done by the sensors. The DHT11 sensor measures temperature and humidity. The LDR sensor detects ambient light levels . The soil moisture sensor measures moisture level of the soil. Processing and Decision making is done by microcontroller (e.g., Arduino Uno). It processes the sensor data. Based on predefined thresholds, the system decides when to activate actuators. If temperature exceeds a threshold, the relay module triggers the fan kit to reduce heat. If light levels drop below a set limit, the light strap is turned on via the relay. If the soil moisture level falls below the threshold, the relay activates the water pump to ensure proper irrigation. The LCD display continuously shows temperature, humidity. The system is powered by two 9V batteries with a diode IN4007 to prevent reverse current flow. This methodology ensures that the greenhouse operates efficiently, providing a stable environment for plant growth with minimal human intervention.



CIRCUIT DIAGRAM:



SIMULATION FOR LCD:



CODE:

```
// Pin Definitions
const int moisturePin = A0;  // Analog pin connected to the soil moisture sensor
const int pump = 2;          // Pin connected to the water pump
const int temppin = A1;      // Analog pin for temperature sensor
const int fan = 3;           // Pin connected to the cooling fan
const int ldrpin = A2;       // Analog pin for light intensity sensor
const int light = 4;         // Pin connected to the light source

// Thresholds
const int dryThreshold = 600;      // Moisture level threshold
const int tempThreshold = 200;     // Temperature threshold
const int intensityThreshold = 600; // Light intensity threshold

void setup() {
    Serial.begin(9600);

    // Set output pins
    pinMode(pump, OUTPUT);
    pinMode(fan, OUTPUT);
    pinMode(light, OUTPUT);
}

void loop() {
    // Read sensor values
    int moistureLevel = analogRead(moisturePin);
    int tempLevel = analogRead(temppin);
    int intensityLevel = analogRead(ldrpin);

    // Print values to Serial Monitor
    Serial.print("Moisture Level: ");
    Serial.println(moistureLevel);
    Serial.print("Temperature Level: ");
    Serial.println(tempLevel);
```

```
Serial.print("Intensity Level: ");
Serial.println(intensityLevel);

// Control water pump
if (moistureLevel < dryThreshold) {
    digitalWrite(pump, HIGH);
} else {
    digitalWrite(pump, LOW);
}

// Control cooling fan
if (tempLevel > tempThreshold) {
    digitalWrite(fan, HIGH);
} else {
    digitalWrite(fan, LOW);
}

// Control light
if (intensityLevel < intensityThreshold) {
    digitalWrite(light, HIGH);
} else {
    digitalWrite(light, LOW);
}

delay(1000);  // Update every second
```

CODE FOR LCD:

```
#include <Wire.h>
#include <LiquidCrystal_I2C.h>
#include <DHT.h>
#define DHTPIN 3        // DHT22 Data pin connected to D3
#define DHTTYPE DHT22   // Using DHT22 sensor
#define RELAY_PIN 4     // Relay control pin
DHT dht(DHTPIN, DHTTYPE);
LiquidCrystal_I2C lcd(0x27, 16, 2);  // Adjust address if needed (0x3F or 0x27)

void setup() {
    Serial.begin(9600);
    dht.begin();
    lcd.init();
    lcd.backlight();

    pinMode(RELAY_PIN, OUTPUT);
    digitalWrite(RELAY_PIN, LOW); // Start with relay OFF

    lcd.setCursor(0, 0);
    lcd.print("DHT22 & Relay");
    delay(2000);
    lcd.clear();
}

void loop() {
    float temperature = dht.readTemperature();  // Read temperature in Celsius
    float humidity = dht.readHumidity();       // Read humidity

    if (isnan(temperature) || isnan(humidity)) {
        Serial.println("Failed to read from DHT sensor!");
        lcd.setCursor(0, 0);
        lcd.print("Sensor Error!");
        return;
    }
```

```
  // Print to Serial Monitor
  Serial.print("Temp: ");
  Serial.print(temperature);
  Serial.print("°C  Humidity: ");
  Serial.print(humidity);
  Serial.println("%");

  // Display on LCD
  lcd.clear();
  lcd.setCursor(0, 0);
  lcd.print("Temp: ");
  lcd.print(temperature);
  lcd.print("C");

  lcd.setCursor(0, 1);
  lcd.print("Humidity: ");
  lcd.print(humidity);
  lcd.print("%");

  // Relay Control - Turn ON if temp > 30°C, OFF otherwise
  if (temperature > 30) {
    digitalWrite(RELAY_PIN, HIGH);
    Serial.println("Relay ON");
  } else {
    digitalWrite(RELAY_PIN, LOW);
    Serial.println("Relay OFF");
  }

  delay(1000); // Update every 1 seconds
}
```

RESULT

The Smart Greenhouse Automation System successfully regulates temperature, humidity, and lighting using sensors and actuators controlled by an Arduino Uno. The system performs the following functions:

- Temperature and Humidity Monitoring: The DHT11 sensor accurately measures temperature and humidity inside the greenhouse. The data is displayed on an LCD screen for real-time monitoring.
- Automatic Climate Control: A fan or exhaust system is activated via a relay module when the temperature exceeds a threshold. A water pump or misting system is triggered to maintain optimal humidity levels.
- **Soil moisture** levels were tracked, triggering irrigation when necessary to ensure plants received adequate water.
- Lighting Automation: The lighting system adjusts based on ambient light conditions. Ensures proper light exposure for plant growth during low-light hours.
- Power Efficiency and Automation: The system operates automatically with minimal human intervention. Reduces energy consumption by activating devices only when necessary.

CONCLUSION

The Smart Greenhouse Automation System provides an efficient, cost-effective, and automated solution for maintaining optimal plant-growing conditions. It led to cost savings, sustainability, and optimized resource use. The system is scalable and could be expanded for larger greenhouses or more detailed monitoring. Future improvements could involve machine learning for predictive analysis, further enhancing the system's performance. The use of sensors, microcontrollers, and automation techniques ensures:

- Improved plant growth and yield through optimal environmental control.
- Reduced manual labor by automating temperature, humidity, and lighting adjustments.

FUTURE SCOPE:

1. AI and Machine Learning for Smart Decision-Making

- Using AI to predict plant diseases and growth patterns.
- Machine learning algorithms to optimize water, light, and temperature conditions.
- Smart alerts for extreme weather conditions or plant health issues.

2. Advanced Sensors and Automation

- Upgrading from **DHT11 to DHT22** for better temperature and humidity accuracy.
- Soil nutrient and pH sensors to optimize fertilizer use.
- Automated irrigation systems using moisture sensors and AI-based scheduling.

3. Renewable Energy Integration

- Using **solar panels** to power the greenhouse, making it more sustainable.

- Energy-efficient components to reduce power consumption.

4. Integration with Robotics

- Automated **planting and harvesting** using robotic arms.
- Drones for plant health monitoring and pest detection.

5. Expansion to Large-Scale Smart Farming

- Scaling up Arduino-based automation to large commercial greenhouses.
- Implementing **wireless mesh networks** for multiple greenhouse control.